

```
> install.packages("earth")
Installing package into '/home/shakthi/R/x86_64-pc-linux-gnu-library/4.1'
...
* building package indices
* testing if installed package can be loaded from temporary location
* checking absolute paths in shared objects and dynamic libraries
* testing if installed package can be loaded from final location
* testing if installed package keeps a record of temporary installation path
DONE (earth)

> library(earth)
Loading required package: Formula
Loading required package: plotmo
Loading required package: plotrix
Loading required package: TeachingDemos
```

The *mda* package provides mixture and flexible discriminant analysis, MARS, and vector-response smoothing splines. You can install and load the *mda* library using the following commands:

```
> install.packages("mda")

Installing package into '/home/shakthi/R/x86_64-pc-linux-gnu-library/4.1'
...
** checking absolute paths in shared objects and dynamic libraries
* testing if installed package can be loaded from final location
* testing if installed package keeps a record of temporary installation path
DONE (mda)

> library(mda)
Loading required package: class
Loaded mda 0.5-4
```

Glass data set

The *mlbench* package contains numerous machine learning benchmark problems and data sets from the UCI repository. Let us use the Glass data set that contains 214 observations of chemical analysis of seven different types of glass.

```
> library(mlbench)
> data(Glass)
```

The *set.seed()* function is used to reset the random number generation. You can then take a sample data from the Glass data set for training and test validation as shown below:

```
> set.seed(103)
> train <- sample(1:dim(Glass)[1], 150)

> data <- Glass[ train, ]
> test <- Glass[-train, ]
```

The structure of the Glass data is given below:

```
> str(data)
'data.frame': 150 obs. of 10 variables:
 $ RI : num 1.52 1.52 1.52 1.52 1.52 ...
 $ Na : num 13.6 13.7 13.1 13.4 13.7 ...
 $ Mg : num 3.87 3.84 3.45 3.49 0 3.5 3.26 3.61 0 3.48 ...
 $ Al : num 1.27 0.72 1.76 1.52 0.56 1.48 2.22 1.54 3.04
1.35 ...
 $ Si : num 72 71.8 72.5 72.7 74.5 ...
 $ K : num 0.54 0.17 0.6 0.67 0 0.6 1.46 0.66 6.21 0.64
...
 $ Ca : num 8.32 9.74 8.38 8.08 10.99 ...
 $ Ba : num 0 0 0 0 0 1.63 0 0 0 ...
 $ Fe : num 0.32 0 0.17 0.1 0 0 0 0 0 ...
 $ Type: Factor w/ 6 levels "1","2","3","5",...: 2 1 3 2 2 2 6
2 4 1 ...
```

The *densityplot()* of the training data set is shown in Figure 1.

bagFDA()

The *bagFDA()* function is a wrapper function that uses multivariate adaptive regression splines (MARS) to implement flexible discriminant analysis (FDA). It accepts the following arguments:

Argument	Description
x	A data frame or matrix of examples
y	A matrix or data frame of outcomes
weights	Weights for examples (Default: 1)
B	Number of bootstrap samples
keepX	Boolean to retain original training data
formula	Represented as $y \sim x_1 + x_2 + \dots$
data	Data frame from which variables are fetched
subset	An index vector listing the samples to be used
na.action	A function to be applied if data has NA